

MCSR Database Project – Documentation

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1. External Libraries & Tools Used

Programming Language

- **Java 21**

GUI Framework

- **Java Swing** (standard JDK components)

Database & JDBC

- **MySQL Community Server**
- **MySQL Connector/J**
Used to connect the Java application to the MySQL database.

Connection Pooling

- **HikariCP**

Logging

- **slf4j-api**
- **slf4j-simple**
Lightweight logging backend for development and debugging.

Database Administration Tools

- **DBeaver**
Used to browse tables, run queries, and export/import datasets.
- **MySQL Command Line Client**
Used for schema creation and running SQL scripts.

Diagramming Tool

- **draw.io**
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2. Installation Instructions

These instructions assume the grader is running the project on a machine with **Java 21+** and **MySQL** installed.

2.1 — Set up the MySQL Database

1. Start the MySQL server.
2. Open a SQL client.
3. Run the schema file:
 - a. `SOURCE db_sql/schema.sql;`
4. Import dataset from exported tables:
 - a. `SOURCE db_exports/users_export.sql;`
 - b. `SOURCE db_exports/matches_export.sql;`
 - c. `SOURCE db_exports/user_achievements_export.sql;`
 - d. etc.
5. Confirm all tables contain data:
 - a. `USE mcsr ;`
 - b. `SHOW TABLES ;`

```
c. SELECT COUNT(*) FROM users;
```

2.2 — Configure Database Credentials

Next to the executable JAR, there is a file named `config.properties`

Default contents:

```
db.url=jdbc:mysql://localhost:3306/mcsr  
db.user=root  
db.password=
```

Modify these if you use a different username, password, or port.

2.3 — Run the Application

Navigate to the `gui_executable/` directory in a terminal:

```
java -jar mcsr_database.jar
```

If the database is running and the credentials are correct, the main application window will appear. It may take a couple of minutes to load.

3. Usage Instructions

The application contains six primary screens:

1. **Players**
2. **Player Profile**
3. **Favorites**

4. **Matches**
5. **Compare**
6. **Leaderboard**

General Feature (applies to all tables):

- Click any **column header** to sort by that column.
 - First click = ascending
 - Second click = descending
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3.1 — Players Tab

This is the default screen when the application opens.

Features:

- **Search by nickname:**
Type any part of a nickname and click **Search**.
 - A table lists all matching players.
 - **View Profile:**
 - Click a row
 - Then click **View Profile**
Opens the selected player's Player Profile screen.
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3.2 — Players Profile

Features:

- **Season selector:** Choose a specific season or "All Time" to update statistics.
 - **Favorite toggle:**
 - Click to add the player to your Favorites list.
 - Click again to remove them.
 - Displays detailed season stats and lifetime stats.
 - Allows navigation back to Player Search or to other tabs.
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3.3 — Favorites Tab

Features:

- **Refresh button:** Updates favorites list by adding newly selected favorites and removing unfavorited profiles
 - **Open Profile:**
Select a player row→ click **Open Profile**.
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3.4 — Matches Tab

Features:

- Enter a season number to filter matches.
 - Or leave the field blank and click **Load Matches** to load matches from all seasons.
 - Displays match metadata such as type, category, season, result, time, forfeits, etc.
 - Sorting helps isolate unusual runs or specific time ranges.
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3.5 — Compare Tab

Features:

1. Select Player A:

- Enter nickname in **Player A Nick**
- Click **Search A**
- Choose the correct player from the dropdown

2. Select Player B:

- Same workflow as Player A, using the Player B fields

3. Season field:

- Enter a season number for season-specific comparison
- Or leave blank for all-time comparison

4. Compare button:

Generates a detailed statistical comparison between the two selected players.

3.6 — Leaderboard Tab

Contains **four sub-tabs**:

1. Elo Leaderboard

Ranks all players by Elo rating.

2. Fastest Time

Shows the single fastest valid completion time per player.

3. Fastest Average

- Only includes players with > 100 matches
- Season can be selected.

4. Best Win%

- Also season-selectable.
- Only includes players with > 100 matches

Shared feature: Sorting by columns allows quickly identifying top players.

4. Data Sources

Player and match data were obtained from the **MCSR Ranked API** (Minecraft Speedrunning Ranked), then exported and normalized into the project's MySQL schema.

The dataset was transformed into 3NF, using associative tables such as:

- `user_achievements`
- `user_achievement_data`

Foreign keys and multi-valued attributes were resolved through relational decomposition.

5. Internal File Behavior

config.properties

- Stored next to the JAR
- Specifies database URL, username, and password
- Loaded at runtime by `DbManager`

Favorites Storage

- Favorites are stored locally through the `FavoritesManager`